Plant products: starch, lipids and proteins

9B Plant growth · Lesson 4 of 7

Date _____ Class _____

Today I can

- ☐ I can describe how plants use glucose to make starch, cellulose, lipids and proteins, and explain why nitrates are needed.
- ☐ I can test a leaf for starch and explain what happens when a seed germinates.

What happens today

1. Do Now: potato and iodine
2. Learn: Carbohydrates: starch and cellulose · Lipids and proteins · A seed germinates · and more
3. Practical: Testing a variegated leaf for starch
4. Challenge, then the exit ticket

Student Book pages 26 to 27 · Workbook pages 19

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
starch	a polymer made of many glucose molecules joined together; a plant's energy store
polymer	a long molecule made of many small molecules joined in a chain
cellulose	a polymer of glucose used to make plant cell walls
lipid	a fat or oil; insoluble in water
protein	a polymer made of amino acids; enzymes are proteins
amino acid	a small molecule that proteins are built from; needs nitrogen
enzyme	a protein that speeds up a reaction in a living thing
nitrate	a mineral salt that supplies nitrogen for making amino acids
germination	when a seed starts to grow into a seedling
iodine solution	an orange-brown liquid that turns blue-black with starch
Benedict's solution	a blue liquid that turns orange-red when heated with glucose

Quick Review

potato and iodine

1 What colour change would you see if you added iodine solution to a slice of potato?

2 Which substance in the potato causes this change?

3 Which substance made in photosynthesis was used to make it?

Core 1 of 2

Everyone does these. Answer on the card or under it.

1 Practical: testing a variegated leaf for starch.

a) Draw the leaf before the test and after adding iodine. Label the green and white parts on the first drawing, and the blue-black parts on the second.

Before the test	After adding iodine

b) Name the solution used to test for starch and the colour it turns when starch is present.

Core 2 of 2

Everyone does these. Answer on the card or under it.

1 *continued*

c) Explain why only some parts of the leaf changed colour.

2 Complete the sentences. Use the word bank.

Glucose molecules made in photosynthesis are joined together to make a polymer called _____.

Starch is an energy store. It is found in the chloroplasts, in storage organs such as _____, and in seeds.

Sugars are also used to make _____, which builds plant cell walls, and _____ such as the oils in seeds.

3 Match each substance to one of its uses in a plant. Draw a line.

starch	speed up reactions such as photosynthesis and respiration
cellulose	energy store in seeds and potatoes
lipids	build cell walls
proteins	built from amino acids using nitrogen from nitrates
enzymes	make the waterproof cuticle and cell membranes

Challenge 1 of 2

When Core is finished. Full sentences.

C1 A plant was left in the dark for 24 hours. A metal foil stencil was then attached to one leaf and the plant was left in bright light for a day. The leaf was then tested for starch.

a) Explain why the plant was left in the dark first.

b) Predict the result of the starch test and explain it.

Challenge 2 of 2

When Core is finished. Full sentences.

C1 *continued*

c) Explain why a metal foil stencil was used and not a clear plastic one.

C2 Avocado fruit flesh is rich in lipids. Avocado seeds also contain lipids. Explain why the plant puts lipids in each place. The two reasons are different.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Which solution tests for starch, and what colour does it turn?

2 Why do seeds need a store of starch?

3 What are enzymes made of, and which mineral salt do plants need to make them?

Today I can: Goal 1 ☐ Goal 2 ☐